Practical No. 10

Create, Rename and Delete a File using Unix/Linux Commands

1. AIM

To implement creation, renaming and deletion of a file using Unix/Linux commands, and to verify each operation.

2. REQUIREMENTS

- Operating System: Ubuntu / Linux
- Linux terminal
- Text editor: Nano or any editor (optional)
- Hardware: computer, keyboard, monitor

3. SHORT THEORY

Unix/Linux manages files using commands typed in the terminal.

Command	Purpose
`touch`	Creates an empty file if it does not exist (if it exists, it only updates its timestamp).
`mv`	Renames a file, or moves it to another location.
`rm`	Removes (deletes) a file. It is permanent: there is no recycle bin.
`ls`	Lists files and directories.
`cat`	Displays the contents of a file.

4. COMMANDS

```
# 1. Create a file
touch student.txt

# 2. Display files
ls

# 3. Rename the file
mv student.txt college.txt

# 4. Verify renamed file
ls

# 5. Delete the file
rm college.txt

# 6. Verify deletion
ls
```

5. HOW TO EXECUTE (step by step)

Command / Key	What it does
`Ctrl + Alt + T`	Open the Ubuntu terminal.
`mkdir practical10`	Make a fresh folder so that ls shows only our file. (Optional but recommended.)
`cd practical10`	Go inside that folder.

Command / Key	What it does
`touch student.txt`	Creates the empty file student.txt.
`ls`	Shows student.txt. Proves the file was created.
`mv student.txt college.txt`	Renames student.txt to college.txt.
`ls`	Shows college.txt only. Proves it was renamed.
`rm college.txt`	Deletes the file.
`ls`	Shows nothing. Proves the file was deleted.

6. LINE-BY-LINE EXPLANATION

Command	Meaning
`# 1. Create a file`	A comment. The # symbol means the line is ignored; it is only a note.
`touch student.txt`	touch = command, student.txt = file name. Creates an empty file with that name.
`ls`	Lists the files in the current folder. Used to verify.
`mv student.txt college.txt`	mv = move. First name = source (old name), second = destination (new name). Same folder, new name, so it works as rename.
`rm college.txt`	rm = remove. Permanently deletes college.txt.
`ls (last)`	Confirms the file is gone.

7. MEANING OF SYMBOLS

Symbol	Meaning
`#`	Comment line, ignored by the shell.
`$`	The shell prompt. It is shown by the terminal; do not type it.
`.txt`	File extension (text file). Just part of the name.
`~`	Shortcut for your home folder.
`. and ..`	Current folder, and parent folder.

8. EXPECTED OUTPUT

```
$ touch student.txt
$ ls
student.txt

$ mv student.txt college.txt
$ ls
college.txt

$ rm college.txt
$ ls
$
```

After deleting, ls prints nothing because college.txt was the only file in the folder. That is why a fresh folder is used.

9. EXTRA COMMANDS (to show more knowledge)

Command	Meaning
`echo "Hello" > student.txt`	Creates the file with text inside (> sends output into the file).
`cat student.txt`	Shows the text inside the file.
`ls -l`	Long listing: shows permissions, size and date.
`rm -i college.txt`	Asks for confirmation (y/n) before deleting.
`mv college.txt ~/Documents/`	Moves the file to another folder.
`cp student.txt backup.txt`	Copies a file.

10. CONCLUSION

Thus, the creation, renaming and deletion of a file were successfully performed using the Unix/Linux commands touch, mv and rm, and each operation was verified using ls.

11. QUICK VIVA ANSWERS

- Create an empty file: touch.
- Rename a file: mv.
- Delete a file: rm.
- Purpose of ls: lists files and directories.
- cat: displays the contents of a file.
- Can mv move a file to another directory? Yes, e.g. mv file.txt ~/Documents/.
- Can a deleted file be recovered? Not with rm; it is permanent (no recycle bin).